

CLASS 10 · CBSE · NCERT MATHEMATICS · PYQ

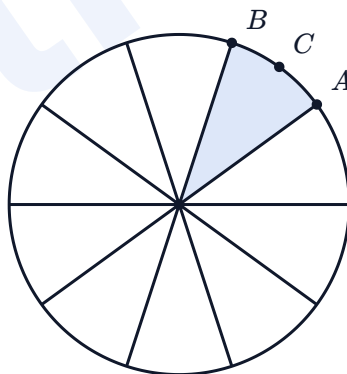
Chapter 11: Areas Related to Circles

Previous year board questions, topic-wise: 4 questions

The tag after each question gives the board examination year and the marks. Solutions are in a separate PDF with the same question numbers.

1 Sector and segment of a circle

1. A sector with a central angle of 72° has an area of 40π square units. Find the radius of the circle. **CBSE 2025, 1 mark**
2. A wire 20 cm long is bent into an arc of a circle of radius $\frac{60}{\pi}$ cm. What angle does the arc subtend at the centre? **CBSE 2025, 1 mark**
3. A circular ornament of diameter 35 mm is made from silver wire. Wire also forms 5 diameters, which cut the circle into 10 equal sectors. A and B are the ends of one sector on the circle and C lies on the arc between them. Take $\pi = \frac{22}{7}$. (i) Find the central angle of each sector. (ii) Find the length of the arc ACB . (iii) Find the area of each sector, OR the total length of wire used. **CBSE 2025, 4 marks**



4. In a circle of radius 21 cm, an arc subtends 60° at the centre. Find (i) the length of the arc (ii) the area of the minor segment cut off by the chord of this arc. **CBSE 2024, 5 marks**

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